1.4.4 ▶ ▶ ▶ Common Moment of Inertias

There are some common moment of inertias that should be memorized to increase speed while solving problems. However, it is also important to have the mathematical flexibility to be able to derive these expressions, especially for Physics C and above. All of these moment of inertias will be taken about an axis which passes through the center of mass[8], although one can find the moment of inertia about other axes with clever use of geometry paired with parallel and perpendicular axis theorem. To derive all of these, we will use the integral definition of moment of inertia, that is

$$I = \int r^2 dm. \tag{1.62}$$

Where the mass element dm is a distance r from the axis. To find dm, we consider a small portion of the object, then use the condition of uniform mass to get dm in terms of an integration variable. For more complicated shapes, we first find the moment of inertia of a small slice and integrate over the entire object.

▶ Starting with the simplest object, a ring of mass M and radius R. Start by considering a small section of the ring of length $Rd\theta$ with mass dm a distance R from the center. Using the condition of uniform mass density gives

$$\frac{dm}{Rd\theta} = \frac{M}{2\pi R}$$
$$dm = \frac{M}{2\pi} d\theta.$$

Plugging into the definition of moment of inertia and integrating yields

$$\begin{aligned} I &= \int_0^{2\pi} R^2 \frac{M}{2\pi} d\theta \\ &= \frac{MR^2}{2\pi} \cdot 2\pi \\ &= \boxed{MR^2} \end{aligned} \tag{1.63}$$

Note that this formula has no dependency on the height of the ring.

▶ Next is a disk of mass M and radius R. Consider a small ring of width dr and mass dm a distance r from the center of the disk. Using the condition of uniform mass density gives

$$\frac{dm}{2\pi r dr} = \frac{M}{\pi R^2}$$
$$dm = \frac{2M}{R^2} r dr.$$

Using eq. 1.62, the moment of inertia of this small ring is

$$dI = dmr^2 = \frac{2M}{R^2} r^3 dr.$$

[8] Which is also the geometrical center since we will only consider uniform mass here.

Integrating over the entire object gives

$$\begin{aligned} I &= \int_0^R \frac{2M}{R^2} r^3 dr \\ &= \frac{2M}{R^2} \cdot \frac{R^4}{4} \\ &= \boxed{\frac{1}{2} M R^2} \end{aligned} \tag{1.64}$$

Notice that this formula has no dependency on the height of the disk.

▶ Next is a rod of mass M and length L centered at the origin. Consider a small section of length dx and mass dm. Using the condition of uniform mass density gives

$$\begin{aligned} \frac{dm}{dx} &= \frac{M}{L} \\ dm &= \frac{M}{L} dx \end{aligned}$$

Using the definition of moment of inertia gives

$$\begin{aligned} I &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{M}{L} x^2 dx \\ &= \frac{M}{L} \cdot \frac{1}{3} \left(\frac{L^3}{8} + \frac{L^3}{8} \right) \\ &= \boxed{\frac{1}{12} M L^2} \end{aligned} \tag{1.65}$$

▶ Next is a spherical shell of mass M and radius R. To find the moment of inertia of this sphere, we can slice it into small rings at an angle of θ from the vertical and with width $Rd\theta$. These rings have mass dm and radius $R \sin\theta$.

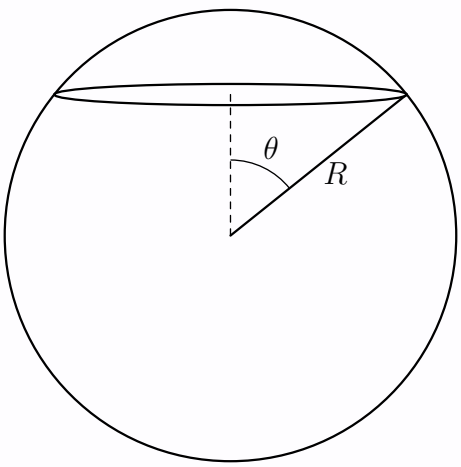

Fig. 1.12

Using the condition of uniform mass density gives

$$\frac{dm}{2\pi R\sin\theta \cdot Rd\theta} = \frac{M}{4\pi R^2}$$

$$dm = \frac{1}{2}M\sin\theta d\theta$$

Using our result from eq. 1.62, the moment of inertia for this small ring is

$$\begin{aligned} dI &= dm \cdot (R\sin\theta)^2 \\ &= \frac{1}{2}MR^2\sin^3\theta \end{aligned}$$

Integrating yields

$$\begin{aligned} I &= \frac{1}{2} \cdot \int_0^\pi \sin^3\theta d\theta \\ &= \frac{1}{2} \cdot \int_0^\pi \sin\theta(1-\cos^2\theta)d\theta \\ &= \frac{1}{2} \cdot \left[-\cos\theta + \frac{1}{3}\cos^3\theta\right]_0^\pi \\ &= \boxed{\frac{2}{3}MR^2} \end{aligned} \tag{1.66}$$

► Lastly, let us consider a solid sphere of mass M and radius R. Let us further consider small spherical shells of radius r and width dr. Using the condition of uniform mass density gives

$$\frac{dm}{4\pi r^2 dr} = \frac{M}{\frac{4}{3}\pi R^3}$$

$$dm = 3\frac{M}{R^3}r^2 dr$$

Using the result from eq. 1.65, the moment of inertia of each spherical shell dI is

$$dI = \frac{2}{3}dmr^2 = 2\frac{M}{R^3}r^4 dr.$$

Lastly, integrating yields

$$\begin{aligned} I &= 2\frac{M}{R^3}\int_0^R r^4 dr \\ &= \boxed{\frac{2}{5}MR^2} \end{aligned} \tag{1.67}$$

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